INCH POUND

MIL-DTL-46874A(MI)

6 JUNE 2018

SUPERSEDING

MIL-S-46874(MI)

14 December 1973

MILITARY SPECIFICATION

SEAL, RF AND PRESSURE

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 Scope. This specification covers the requirements for strips of material consisting of knitted Monel metal bonded to closed cell chloroprene rubber, which functions both as an RF and pressure seal.

1.2 Classification. The strips are classified by dash numbers in accordance with cross-sectional dimensions shown in Figure 1 and Table I.

2. APPLICABLE DOCUMENTS

2.1 General. The document listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3,4, or 5 of this standard, whether or not they are listed.

2.2. Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army Aviation and Missile Command, ATTN: RDMR-SET, Redstone Arsenal, AL 35898-5000 or Hattie.j.Strong.civ@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5330

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FEDERAL SPECIFICATIONS

QQ-N-281	Nickel-Copper-Alloy (Monel and R-Monel) bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	Marking for Shipment and Storage

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

QUALIFIED PRODUCTS LIST

QPL-MMM-A-1617	Adhesive, Rubber Base, General Purpose

(Copies of this QPL are available online at http://qpdocs.dla.mil.)

2.3 Non-Government publications. The following document of the exact revision listed below form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM D429	Rubber Property –Adhesion To Rigd Substrates
ASTM D6576	Rubber, Flexible Cellular, Chemically Blown

(Copies of this document are available from www.astm.org)

3. REQUIREMENTS

3.1 Preproduction sample. When specified by the procuring activity (see 6.2), a preproduction sample shall be prepared using the same methods proposed for subsequent production, and shall be subjected to preproduction inspection (see 4.3).

3.2 General. The material shall consist of knitted Monel wire strips bonded to strips of cellular rubber.

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3.3 Metal.

3.3.1 Wire. The wire shall be Monel conforming to the requirements of QQ- N-281, Class A, and shall have a diameter of .0045 inch.

3.3.2 Compression of wire. Upon application of 20 plus or minus 0.5 pounds per square inch (psi) load, the knitted wire shall compress 17 to 25 percent when tested in accordance with 4.5.1.

3.4 Rubber. The rubber shall conform to the requirements of ASTM- D6576, Type II, Grade A, Condition Medium.

3.5 Adhesive.

3.5.1 Conformance. The adhesive used to bond the wire to the rubber shall conform to the requirements of QPL-MMM-A-1617, Type II.

3.5.2 Thickness. The thickness of the adhesive film between the rubber and the metal shall be .010 inch.

3.5.3 Bond. The bond between the rubber and metal shall pass the test specified in 4.5.2.

3.6 Shape, dimensions, and tolerances.

3.6.1 Shape and dimensions.

3.6.1.1 Specifications. The material shall have the cross-sectional shape and dimensions outlined in Figure 1 and Table I as specified for each dash number.

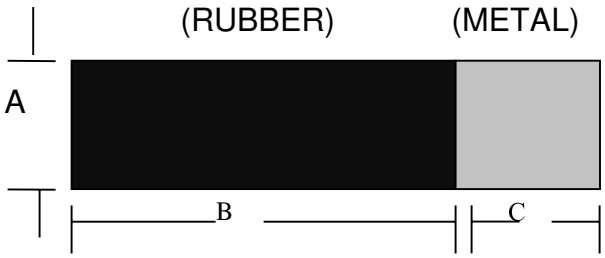


Figure 1

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Table I Dimensions

Specification Number	Dimension, Inch A	B	C
MIL-DTL-46874-1		.25	.25
MIL-DTL-46874-2		.50	.25
MIL-DTL-46874-3		.375	.125
MIL-DTL-46874-4		.50	.50
MIL-DTL-46874-5	.125	.50	.125
MIL-DTL-46874-6		.25	.125
MIL-DTL-46874-7		.19	.125
MIL-DTL-46874-8		.19	.19
MIL-DTL-46874-9		.125	.125
MIL-DTL-46874-10		.75	.125

3.6.1.2 Length. The length shall be as specified in the procurement document.

3.6.2 Tolerances. Dimensional tolerances shall be in accordance with Table II.

Table II. Tolerances

	Dimension (Inch)	Tolerance (Inch) Rubber	Metal
	.06 through .125	+.03, - .015	+.015, -0
Over	.125 through .19	±.03	+.015, -0
Over	.19 through .375	±.03	+.03, -0
Over	.375 through .50	±.06	+.05, -0
Over	.50 through .75	±.06	--

3.7 Product identification.

3.7.1 Information to be shown. Unless otherwise specified in a procuring document, each separate sheet and strip of material shall have a tag attached showing the following:

a. MIL-DTL-46874 and appropriate dash number.

b. Vendor's name and code.

c. Manufacturer's material designation

d. Date of manufacture (year and quarter).

3.7.2 Marking. Marking shall be in accordance with the requirements of MIL-STD-129.

3.8 Workmanship. The material shall be of high quality and uniformity and shall be free of contamination, unevenness, tears, cuts, abnormal holes, and other

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defects. Workmanship shall be such as to ensure a product which is in conformance with this specification.

3.9 Marking. Marking shall be in accordance with MIL-STD-129 and shall include but shall not necessarily be limited to the following:

a. MIL-DTL-46874.

b. Vendor's name.

c. Manufacturer's material designation.

d. Quantity and dimensions.

e. Other special instructions.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.2 Classification and tests. The testing of the material shall be classified as follows:

4.2.1 Preproduction inspection (4.3).

4.2.2 Quality conformance inspection (4.4).

4.3 Preproduction inspection. Preproduction inspection shall consist of all examinations and tests specified herein and shall be conducted only on preproduction samples. The preproduction samples shall be prepared using the same methods to be used for the manufacture of all subsequent production material.

4.4 Quality conformance inspection.

4.4.1 Inspections. Quality conformance inspection tests shall be the examination and tests required for each production lot. A production lot shall consist of finished material made on any single day from individual lots or rubber and metal components.

4.4.2 Sampling. Two 2 foot samples of each dash number material shall be taken at random from each shipment. Tests shall be run in duplicate using on specimen from each of the selected samples.

4.4.3 Quality conformance tests. Quality conformance tests shall include

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checking for dimensional requirements in accordance with 3.6 and comparison of the material with an approved sample to determine similarity of appearance and adequate workmanship in accordance with 3.9.

4.5 Test methods.

4.5.1 Metal compression. In order to determine conformance with 3.3.2, any conventional method may be used.

4.5.2 Adhesion. In order to determine conformance with 3.5.3, the bond between the rubber and metal shall be tested in accordance with Method 8031 or ASTM-D429. Rupture of the rubber and no adhesive failure shall be regarded as a satisfactory bond.

5. PREPARATION FOR DELIVERY

5.1 Packaging. For acquisition purposes, the packaging requirement shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contract or personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This material is intended for use as a compressible gasketing where both RF and pressure sealing are required.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether preproduction samples and test reports are required.

c. Special packaging and packing instructions, if required.

d. Other special instructions.

6.3 Level C. Unless otherwise specified in 6.2 each cross-sectional size of material should be packed separately. The material should not be folded or compressed. Packaging should conform to good commercial practice and should protect the material from contamination, damage, and change of shape during shipment and storage and should assure safe delivery and acceptancy at destination.

6.4 Standard test method. Due to lack of a standard test method for

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determining RF energy attenuation properties there are no such requirements in this specification. It is therefore recommended that the material be tested in the equipment where it is intended to be used, to assure that it is satisfactory.

6.5 Recommendation. It is recommended that the rubber component of the product be examined three years after cure date. If the rubber meets the compression deflection requirements and shows no evidence of cracks on folding, it is suggested that it be considered satisfactory for use.

6.6 Subject term (key word) listing.

Chloroprene Rubber
Closed Sell
Knitted Monel Metal

6.7 Change from previous issue. This document was used to incorporate changes to replaced cancel documents used as references.

CONCLUDING MATERIALS

Custodian:
ARMY-MI

Preparing Activity:
ARMY-MI

(5330-2018-006)

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